REDBIRD FLIGHT SIMULATIONS

QUALIFICATION AND APPROVAL GUIDE (QAG)

CRV ADVANCED AVIATION TRAINING DEVICE
AMENDED QUALIFICATION AND APPROVAL GUIDE
VERSION 2.1B



ADVANCED AVIATION TRAINING DEVICE

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FAA APPROVED QAG
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February 22, 2019

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LOG OF REVISIONS

Description of Changes	Version	Affected Pages	Edited By
Original	1.0	All	CG
Added Log of Revisions	2.0	2	WJ
Updated Section 1 Device Overview	2.0	3	WJ
Moved Figures and Photos to Section 2	2.0	5-15	WJ
Updated Panel images Figure 1 and Figure 2	2.0	5	WJ
Added Visual Displays – Type 2 Figure 6.4	2.0	16	WJ
Moved Hardware and Software Components to Section 3	2.0	18,19	WJ
Added Win10, Visual Display Type 2, Navigator to Table 1	2.0	18,19	WJ
Moved Statements of Compatibility to Section 3	2.0	20	WJ
Moved Statement of Compliance to Section 4	2.0	21-23	WJ
Updated Display Requirements (1) and (2)	2.0	22	WJ
Added Performance table to Section 4	2.0	23	WJ
Edited redundant information from aircraft configuration	2.0	25	WJ
Added new configuration for approval	2.0	26	WJ
Added Section 6 Visual System details	2.0	27,28	WJ
Renamed Section 1 to Compliance Statement	2.0a	3	WJ
Renamed Section 2 Aviation Training Device (ATD) Description and Pictures	2.0a	4	WJ
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Added Figure 6.6 to Section 2	2.0a	6	WJ
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Added Control Loading (Dual) rudder pedals to Section 3	2.1	11	WJ
Updated Compliance Statement to comply with AC61-136B	2.1	3	WJ
Added display types 3 & 4 to Component List	2.1	11	WJ
Updated to comply with AC61-136B	2.1	12 - 17	WJ
Moved configurations added in 2.0B to Previously Approved	2.1	18, 19	WJ
Updated description to include dimensions for larger monitors.	2.1	21	WJ
Added Figures 6.7 and 6.8	2.1	24	WJ
Replaced IS Figure 6.1 with larger image depicting aircraft with airways	2.1A	5	WJ
Added Grand Caravan EX	2.1A	20, 21	WJ
Updated Section 7 for AC61-136B compliance	2.1A	26-29	WJ
Updated text for C3.1.9	2.1B	17	WJ
Updated reference from SD to non-motion	2.1B	10	WJ
Corrected typo referencing FMX, MCX, SD, LD	2.1B	14	WJ
Updated Figure descriptions to include monitor size	2.1B	24, 25	WJ
Enlarged Figure 6.1	2.1B	5	WJ
Replaced Figure 6.2 and 6.6	2.1B	6	WJ

SECTION 1: COMPLIANCE STATEMENT

This Qualification and Approval Guide (QAG) includes a detailed description of all components, functions, and capabilities of the Redbird model CRV. This includes any optional configurations with quality color pictures and diagrams. The required information as described in the current advisory circular, AC 61-136B / FAA Approval of Aviation Training Devices (ATD) and Their Use for Training and Experience, required for FAA approval, is contained within this document. This includes listing all the required qualifying items, functions, and capabilities. This QAG is used to clearly describe and verify the functionality of this aviation training device platform confirming its suitability for airman training and experience. A valid FAA Letter of Authorization (LOA) specifying the ATD allowances must also accompany the training device when utilized for airman training or experience requirements as specified in 14 CFR §61 or 141.

This ATD must continue to maintain its performance and function without degradation. The operator of this device is expected to maintain its condition and functionally when used for airmen training or experience requirements. Only the configurations approved for this model, as described within this QAG, can be utilized when satisfying FAA experience or training requirements. Any additions, changes, or revisions to the model or to the available configurations described within this document must be approved in writing by The General Aviation and Commercial Division, AFS-800.

Any questions concerning FAA approval of ATDs should be directed to the General Aviation and Commercial Division, AFS-800.

SECTION 2: AVIATION TRAINING DEVICE (ATD) DESCRIPTION AND PICTURES

The Redbird model CRV is based on the dimensions and layout of a Cessna Grand Caravan (C208B) aircraft. This closely represents the overall functionality, performance, avionics, and instrumentation. The platform consists of a cockpit section, instructor's control station, visual display system and an audio system. It incorporates a combination of hardware and software components that is assembled and checked by Redbird Flight Simulations. All hardware elements are permanently installed and designed so the cockpit has the appearance and feel of an actual aircraft. From the pilot's seated position, there are no computer hardware elements such as keyboards, pointing devices, etc. for his or her use.

The CRV provides realistic and true-to-scale cockpit design, avionics, and reliable hardware/software performance. This platform also provides an effective training environment for student and certificated pilots. This includes the capability of practicing scenario based flight training events, simulated equipment failure and emergency procedures, pilot evaluations, instrument procedures/ experience, and facilitating increased pilot proficiency overall.

The Redbird CRV is a versatile and affordable device that has been designed to represent a Cessna Grand Caravan. It is equipped with the following notable features:

- 3-axis electric motion platform providing pitch, roll and yaw motions
- Wrap-around exterior visuals provided by 6 or 8 LCD screens
- Dual pilot controls including a 2-axis control-loaded yoke and interconnected rudder pedals
- Realistic switches, buttons, knobs, circuit breakers and other cockpit controls that are designed to match the aircraft wherever possible
- An interchangeable instrument panel to allow development of future configurations
- Closed Circuit intercom system, allowing for communication between the pilot, co-pilot and instructors using standard aviation headsets
- A portable instructors station, allowing the instructor to operate from inside or outside the simulator
- Optional supplemental oxygen system for the pilot and/or co-pilot

Configuration Components

Instructor's Station

The Redbird Instructor Station interface is operated through any PC and/or web browser enabled device. The Instructor Station software is available as an installed program on the PC and web-browser based application. PC based Instructor Station can be on either a desktop computer or laptop/mobile PC.



Figure 6.1 Instructors PC/Tablet or Mobile Device

Aircraft Instrument Configurations

The Redbird CRV simulator is available with the traditional analog instruments and the glass panel instrument configuration. Switching between glass and analog configurations only involve changing out the instrument acrylic panel.



Figure 6.2 Assembled Cockpit (Glass)



Figure 6.6 Assembled Cockpit (Analog)

Controls



Figure 2.1 Pilot Yoke



Figure 2.5 Co-Pilot Yoke



Figure 2.2 Pilot Yoke Switches – Left

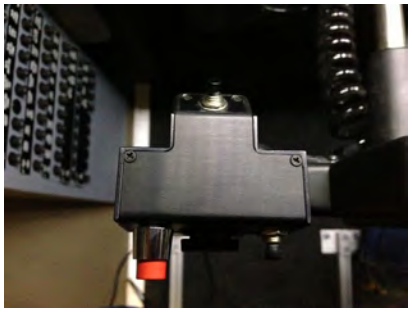


Figure 2.3 Pilot Yoke Switches – Left

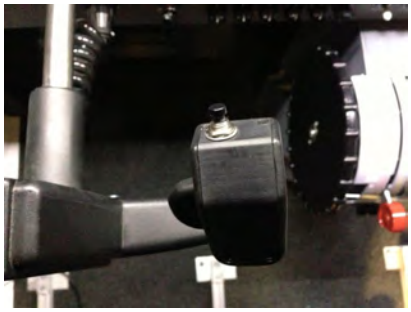


Figure 2.4 Pilot Yoke Switches – Right



Figure 2.6 Co-Pilot Yoke Switches – Left



Figure 2.7 Co-Pilot Yoke Switches – Right

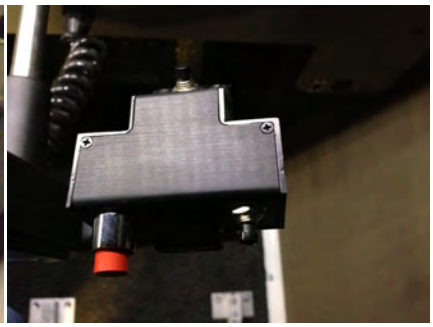


Figure 2.8 Co-Pilot Yoke Switches – Right



Figure 3.1 Throttle Quadrant



Figure 4.1 Overhead Panel



Figure 4.2 Circuit Breaker Panel Pilots Side



Figure 4.4 Left Switch Panel



Figure 4.3 Circuit Breaker Panel and Left Switch Panel



Figure 4.5 Switch Panel Pilots Side 1



Figure 4.6 Lower Switch Panel Pilots Side 2



Figure 4.7 Lower Switch Panel Center



Figure 4.8 Lower Switch Panel Co-Pilots Side

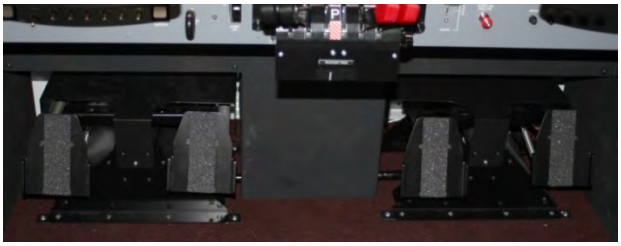


Figure 5.1 Rudder Pedals Pilot and Co-Pilot



Figure 6.5: Motion Platform and Cockpit Enclosure

SECTION 3: TRAINING DEVICE COMPONENTS LIST

Qty	Type	Manufacturer	Name	Description/Function	Configuration
1	Software	Microsoft	Windows 7, or 10	Operating system. (depending on serial number)	All
1	Software	Lockheed Martin	Prepar3D	Simulation engine. (depending on serial number)	All
1	Software	Redbird Flight Simulations	RB Sim	Simulation control and component integration. (depending on serial number)	All
1	Software	Redbird Flight Simulations	FMX Control	Motion system control and component integration. (depending on serial number)	All – Except when configured as non-motion
1	Software	Redbird Flight Simulations	Instructors Station	Environmental, location and failure controls with map, track and glideslope display.	All
1	Software	Redbird Flight Simulations	Navigator	Simulation control and component integration. Environmental, location and failure controls with map, track and glideslope display. (depending on serial number)	All
1	Software	Redbird Flight Simulations	Parrot and Cygnus	Optional software providing ATC and location services	All
1	Software	Mindstar Aviation	Redbird 1000 (CRV) and CRV Gauge	Virtual flight instruments, radios, gauges, indicators, alerts, misc. instruments and logic controls for simulated systems.	See Configuration
1	Software	Mindstar Aviation	Redbird 430/ 530	Virtual GPS and radios	See Configuration
1	Software	Mindstar Aviation	Redbird 850	Virtual MFD	See Configuration
1	Software	Mindstar Aviation	Redbird Autopilot	Virtual Autopilot (Redbird KFC225, GFC700)	See Configuration
1	Software	Mindstar Aviation	Redbird Audio Panel	Virtual Audio Panel (GMA340)	See Configuration
1	Software	Mindstar Aviation	Redbird Transponder	Virtual Transponder (GTX330)	See Configuration
1	Software	Redbird Flight Simulations	Analog Gauges	Virtual Airspeed, Attitude, Altimeter, Turn and Bank, Heading, HSI, VSI, RMI, CDI, and ADF Gauges , Radios	See Configuration
1	Software	Redbird Flight Simulations	Miscellaneous Gauges	Virtual Miscellaneous Gauges	All
1	Software	Mindstar Aviation	Miscellaneous Gauges	Virtual Miscellaneous Gauges	All
1	Software	Flight 1	Compass	Virtual Compass	All
1	Hardware	Redbird Flight Simulations	Simulation Computer	Host computer for flight simulation engine, simulation control software, airplane systems and instruments.	All
1	Hardware	Redbird Flight Simulations	Motion Control Computer	Host computer for FMX Control.	All – Except when configured as non-motion
1	Hardware	Industry Standard	Instructors Station Computer or Wireless Mobile Device	Host computer for Instructors Station.	All
1	Hardware	Redbird Flight Simulations	Motion Platform: Type 1	Gimbaled, steel motion platform with movement in pitch, roll and yaw. Includes all motors, sensors and safety controls.	All – Except when configured as non-motion
1	Hardware	Redbird Flight Simulations	Cockpit Enclosure	Cockpit enclosure to exclude distractions.	All
6	Hardware	22” LCD Industry Standard Monitor	Visual Display - Type 1	Flat Panel displays for exterior views.	All – Varies by configuration. See Figure 6.3
8	Hardware	22” LCD Industry Standard Monitor	Visual Display - Type 2	Optional Flat Panel displays for exterior views.	All – Varies by configuration. See Figure 6.4

6	Hardware	27" LCD Industry Standard Monitor	Visual Display - Type 3	Optional large Flat Panel displays for exterior views.	All – Varies by configuration. See Figure 6.7
8	Hardware	27" LCD Industry Standard Monitor	Visual Display - Type 4	Optional large Flat Panel displays for exterior views.	All – Varies by configuration. See Figure 6.8
3	Hardware	19" LCD Industry Standard Monitor	LCD	Flat Panel displays for virtual instruments.	All
1	Hardware	Redbird Flight Simulations	Yoke (Dual)	Dual, Control loaded pitch and roll controller with switches and buttons for airplane systems operation.	All
1	Hardware	Redbird Flight Simulations	Throttle Quadrant	Single, turbine engine control with throttle lever, propeller control lever, mixture lever, fuel cutoff lever, flap position lever and indicator and pitch, elevator, aileron and rudder trim controls.	All
1	Hardware	Redbird Flight Simulations	Rudder Pedals (Dual)	Pilot and co-pilot rudder control pedals with toe brakes.	All – Varies by configuration
1	Hardware	Redbird Flight Simulations	Rudder Pedals (Dual) – Control Loading	Optional Control loading pilot and co-pilot rudder control pedals with toe brakes.	All – Varies by configuration
1	Hardware	Redbird Flight Simulations	Switch Panel	Lower switch panel with airplane configuration and systems controls.	All
1	Hardware	Redbird Flight Simulations	Left Switch Panel	Left side switch panel with airplane systems	All
1	Hardware	Redbird Flight Simulations	Oxygen System	Optional oxygen system for the pilot and/or copilot	All
1	Hardware	Redbird Flight Simulations	Circuit Breaker Panel	Pilot side circuit breaker panel.	All
1	Hardware	Redbird Flight Simulations	Overhead Control Panel	Overhead control panel.	All
1	Hardware	Redbird Flight Simulations	Instrument Controls Overlay	Flight instruments, radios, airplane configuration and systems controls as required for each configuration.	See Configuration

Table 1: Training Device Component List (Above)

Statements of Compatibility of Software and Hardware

AC 61-136B, Appendix A.6.2

The device and all configurations included in this Qualification and Approval Guide (QAG) meet the requirements for the Compatibility of Software and Hardware.

Microsoft Windows (Operating System)

This is to certify that Microsoft Corporation, the owner and developer of the Windows operating system, has evaluated that their operating system works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software.

Lockheed Martin Prepar3D (Simulation Engine)

This is to certify that Lockheed Martin, the owner and developer of the Prepar3D Simulator Engine, has evaluated that their software application works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software

Mindstar Aviation Redbird Virtual Instrument Suite

This is to certify that Mindstar Aviation, the developer of the Redbird Virtual Instrument Suite, including the Redbird G1000 (CRV), CRV Gauge, Redbird 430/530, Redbird GMA340, Redbird KFC225 Autopilot, Redbird KMD850 GPS, Redbird KR87 ADF, Redbird CTX330, Redbird Radios and Redbird Flight Simulations, the hardware manufacture and component integrator, have demonstrated that the Redbird Virtual Instrument Suite software package is fully compatible with the Redbird Flight Simulation model CRV and all configurations of that model. Mindstar Aviation and Redbird Flight Simulations can assure that the communications/transport data delay is not greater than 300 milliseconds and all analog and digital input signals meet the performance criteria established for the software performance by Redbird Flight Simulations.

The device and all configurations included in this Qualification and Approval Guide (QAG) meet the requirements for the Compatibility of Software and Hardware.

Flight 1 Compass

This is to certify that Flight 1, the owner and developer of the Compass and Redbird Flight Simulations, the hardware manufacturer and component integrator, have demonstrated that the Compass software is fully compatible with the Redbird Flight Simulations model CRV. Redbird Flight Simulations can assure that the communications/transport data delay is not greater than 300 milliseconds and all analog and digital input signals meet the performance criteria established for the software performance by Redbird Flight Simulations.

SECTION 4: AVIATION TRAINING DEVICE (ATD) DESIGN CRITERIA LIST

The following section provides the detailed "word for word" listing and design criteria of each of the required items, functions, and capabilities (listed in AC 61-136B, for BATD requirements Appendix B and the additional AATD items of Appendix C) and operational performance value/scale (as applicable) for each of the functions described for the Redbird CRV.

Basic ATD Requirements List [Appendix B items]

All configurations for this model, as noted, meet AC 61-136B, Appendix B requirements

The Redbird CRV meets the following (General Control Requirements):

B.3.1.1. The aircraft physical flight and associated control systems must be recognizable as to their function and how they are to be manipulated solely from their appearance. These physical flight control systems cannot use interfaces such as a keyboard, mouse, or gaming joystick to control the aircraft in simulated flight.

B.3.1.2. Virtual controls are those controls used to set up certain aspects of the simulation (such as selecting the aircraft configuration, location, weather conditions, etc.) and otherwise program, effect, or pause the training device. These controls are often part of the instructor station or independent computer interface.

B.3.1.3. Except for the initial setup, a keyboard or mouse may not be used to set or position any feature of the ATD flight controls for the maneuvers or training tasks to be accomplished. See the control requirements listed below as applicable to the aircraft model represented. The pilot must be able to operate the controls in the same manner as it would be in the actual aircraft. This includes the landing gear, wing flaps, cowl flaps, carburetor heat, mixture, propeller, and throttle controls appropriate to the aircraft model represented.

B.3.1.4. The physical arrangement, appearance, and operation of controls, instruments, and switches required by this appendix should closely model the aircraft represented. Manufacturers are expected to recreate the appearance, arrangement, operation, and function of realistically placed physical switches and other required controls representative of an aircraft instrument panel that includes at least the following:

- Master/battery;
- Magnetos for each engine (as applicable);
- Alternators or generators for each engine;
- Auxiliary power unit (APU) (if applicable);
- Fuel boost pumps/prime boost pumps for each engine;
- Avionics master;
- Pitot heat; and
- Rotating beacon/strobe, navigation, taxi, and landing lights. Only the software evaluated and approved by the FAA is installed on this computer system.

B.3.1.5. When an FAA-approved ATD is in use, only the software evaluated by the FAA may be loaded for use on that computer system. This does not preclude providing software updates that do not otherwise change the appearance of the systems operation.

The Redbird CRV model meets the following Control Requirements (For Airplane):

B.3.2.1.1 A self-centering displacement yoke or control stick that allows continuous adjustment of pitch and bank.

B.3.2.1.2 Self-centering rudder pedals that allow continuous adjustment of yaw and corresponding reaction in heading and roll.

B.3.2.1.3 Throttle or power control(s) that allows continuous movement from idle to full-power settings and corresponding changes in pitch and yaw, as applicable.

B.3.2.1.4 Mixture/condition, propeller, and throttle/power control(s) as applicable to the M/M of aircraft represented.

B.3.2.1.5 Controls for the following items, as applicable to the category and class of aircraft represented:

- Wing flaps,
- Pitch trim,

- Communication and navigation radios,
- Clock or timer,
- Gear handle (if applicable),
- Transponder,
- Altimeter,
- Carburetor heat (if applicable), and
- Cowl flaps (if applicable).

The Redbird CRV meets the following (Control Input Functionality and Response Criteria):

B.3.3.1 Time from control input to recognizable system response must be without delay (i.e., not appear to lag in any way). “Redbird Flight Simulations, Inc.” verifies that the “Redbird CRV” meets this requirement.

B.3.3.2 The control inputs must be tested by the computer and software program at each startup and displayed as a confirmation message of normal operation or a warning message that the transport delay time or any design parameter is out of tolerance. It should not be possible to continue the training session unless the problem is resolved and all components are functioning properly.

The Redbird CRV meets the following (Display Requirements):

B.3.4.1 The following instruments and indicators must be replicated and properly located as appropriate to the aircraft represented:

B.3.4.1.1 Flight instruments in a standard configuration representing the traditional “round” dial flight instruments. An electronic primary flight display (PFD) with reversionary and backup flight instruments is also acceptable.

B.3.4.1.2 A sensitive altimeter with incremental markings each 20 feet or less, operable throughout the normal operating range of the M/M of aircraft represented.

B.3.4.1.3 A magnetic direction indicator.

B.3.4.1.4 A heading indicator with incremental markings each 5 degrees or less, displayed on a 360 degree circle. Arc segments of less than 360 degrees may be selectively displayed if desired or required, as applicable to the M/M of aircraft represented.

B.3.4.1.5 An airspeed indicator with incremental markings as shown for the M/M aircraft represented; airspeed markings of less than 20 knots need not be displayed.

B.3.4.1.6 A vertical speed indicator (VSI) with incremental markings each 100 feet per minute (fpm) for both climb and descent, for the first 1,000 fpm of climb and descent, and at each 500 fpm climb and descent for the remainder of a minimum ±2,000 fpm total display, or as applicable to the M/M of aircraft being represented.

B.3.4.1.7 A gyroscopic rate-of-turn indicator or equivalent with appropriate markings for a rate of 3 degrees per second turn for left and right turns. If a turn and bank indicator is used, the 3 degrees per second rate index must be inside of the maximum deflection of the indicator.

B.3.4.1.8 A slip and skid indicator with coordination information displayed in the conventional inclinometer format where a coordinated flight condition is indicated with the ball in the center position. A split image triangle indication as appropriate for PFD configurations may be used.

B.3.4.1.9 An attitude indicator with incremental markings each 5 degrees of pitch or less, from 20 degree pitch up to 40 degree pitch down or as applicable to M/M of aircraft represented. Bank angles must be identified at “wings level” and at 10, 20, 30, and 60 degrees of bank (with an optional additional identification at 45 degrees) in left and right banks.

B.3.4.1.10 Engine instruments as applicable to the M/M of aircraft being represented, providing markings for the normal ranges including the minimum and maximum limits.

B.3.4.1.11 A suction gauge or instrument pressure gauge with a display applicable to the aircraft represented.

B.3.4.1.12 A flap setting indicator that displays the current flap setting. Setting indications should be typical of that found in an actual aircraft.

B.3.4.1.13 A pitch trim indicator with a display that shows zero trim and appropriate indices of airplane nose down and airplane nose up trim, as would be found in an aircraft.

B.3.4.1.14 Communication radio(s) with a full range of selectable frequencies displaying the radio frequency in use.

B.3.4.1.15 Navigation radio(s) with a full range of selectable frequencies displaying the frequency in use and capable of replicating both precision and nonprecision instruments, including approach procedures (each with an aural identification feature), and a marker beacon receiver. For example, an instrument landing system (ILS), non-directional radio beacon (NDB), Global Positioning System (GPS), Localizer (LOC) or very high frequency omni-directional range (VOR). Graduated markings as indicated below must be present on each course deviation indicator (CDI) as applicable. The marking should include:

- One-half dot or less for course/glideslope (GS) deviation (i.e., VOR, LOC, or ILS), and
- Five degrees or less for bearing deviation for automatic direction finder (ADF) and radio magnetic indicator (RMI), as applicable.

B.3.4.1.16 A clock with incremental markings for each minute and second, or a timer with a display of minutes and seconds.

B.3.4.1.17 A transponder that displays the current transponder code.

B.3.4.1.18 A fuel quantity indicator(s) that displays the fuel remaining, either in analog or digital format, appropriate for M/M of aircraft represented.

B.3.4.2 All instrument displays listed above must be visible during all flight operations. Allowances can be made for multifunction electronic displays that may not display all instruments simultaneously. All of the displays must provide an image of the instrument that is clear and:

B.3.4.2.1 Does not appear to be out of focus or illegible.

B.3.4.2.2 Does not appear to “jump” or “step” during operation.

B.3.4.2.3 Does not appear with distracting jagged lines or edges.

B.3.4.2.4 Does not appear to lag relative to the action and use of the flight controls.

B.3.4.3 Control inputs should be reflected by the flight instruments in real time and without a perceived delay in action. Display updates must show all changes (within the total range of the replicated instrument) that are equal to or greater than the values stated below:

B.3.4.3.1 Airspeed indicator: change of 5 knots.

B.3.4.3.2 Attitude indicator: change of 2 degrees in pitch and bank.

B.3.4.3.3 Altimeter: change of 10 feet.

B.3.4.3.4 Turn and bank: change of ¼ standard rate turn.

B.3.4.3.5 Heading indicator: change of 2 degrees.

B.3.4.3.6 VSI: change of 100 fpm.

B.3.4.3.7 Tachometer: change of 25 rpm or 2 percent of turbine speed.

B.3.4.3.8 VOR/ILS: change of 1 degree for VOR or ¼ of 1 degree for ILS.

B.3.4.3.9 ADF: change of 2 degrees.

B.3.4.3.10 GPS: change as appropriate for the model of GPS-based navigator represented.

B.3.4.3.11 Clock or timer: change of 1 second.

B.3.4.4 Displays must reflect the dynamic behavior of an actual aircraft (e.g., a VSI reading of 500 fpm must reflect a corresponding movement in altitude, and an increase in power must reflect an increase in the rpm indication or power indicator.)

The Redbird CRV meets the following (Flight Dynamics Requirements):

B.3.5.1 Flight dynamics of the ATD should be comparable to the way the represented training aircraft performs and handles. However, there is no requirement for an ATD to have control loading to exactly replicate any particular aircraft.

B.3.5.2 Aircraft performance parameters (such as maximum speed, cruise speed, stall speed, maximum climb rate, and hovering/sideward/forward/rearward flight) should be comparable to the aircraft being represented. A performance table will need to be included in the QAG for each aircraft configuration for sea level and 5,000 feet using standard atmosphere and gross weight conditions. An alternate performance altitude for 6,000 feet can be used if the manufacturer of that aircraft has a performance chart reflecting that altitude; otherwise the ATD manufacturer will need to interpolate the performance for the chart. Performance at altitude for turboprop or turbojet configurations should reflect 18,000 ft.

B.3.5.3 Aircraft vertical lift component must change as a function of bank comparable to the way the aircraft being represented performs and handles.

B.3.5.4 Changes in flap setting, slat setting, gear position, collective control, or cyclic control must be accompanied by changes in flight dynamics comparable to the way the M/M of aircraft represented performs and handles.

B.3.5.5 The presence and intensity of wind and turbulence must be reflected in the handling and performance qualities of the simulated aircraft and should be comparable to the way the aircraft represented performs and handles.

The Redbird CRV meets the following Instructional Management Requirements:

B.3.6.1 The instructor must be able to pause the system at any time during the training simulation for the purpose of administering instruction or procedural recommendations.

B.3.6.2 If a training session begins with the "aircraft in the air" and ready for the performance of a particular procedural task, the instructor must be able to manipulate the following system parameters independently of the simulation:

- Aircraft geographic location,
- Aircraft heading,
- Aircraft airspeed,
- Aircraft altitude, and
- Wind direction, speed, and turbulence.

B.3.6.3 The system must be capable of recording both a horizontal and vertical track of aircraft movement during the entire training session for later playback and review.

B.3.6.4 The instructor must be able to disable any of the instruments prior to or during a training session and be able to simulate failure of any of the instruments without stopping or freezing the simulation to affect the failure. This includes simulated engine failures and the following aircraft systems failures: alternator or generator, vacuum or pressure pump, pitot static, electronic flight displays, or landing gear or flaps, as appropriate.

B.3.6.5 The ATD must have at least a navigational area database that is local (25 nautical miles (NM)) to the training facility to allow reinforcement of procedures learned during actual flight in that area. All navigational data must be based on procedures as published per 14 CFR part 97.

Advanced ATD Requirements List [Appendix C items]

All configurations, as noted in AC 61-136B, Appendix C meet these additional AATD design criteria items listed.

The Redbird CRV meets the following additional AATD CRITERIA:

C.3.1.1 A realistic shrouded (enclosed) or unshrouded (open) cockpit design and instrument panel arrangement representing a specific model aircraft cockpit.

C.3.1.2 Cockpit knobs, system controls, switches, and/or switch panels in realistic sizes and design appropriate to each intended functions, in the proper position and distance from the pilot's seated position, and representative of the category and class of aircraft being represented.

C.3.1.3 Primary flight and navigation instruments appropriately sized and properly arranged that exhibit neither stepping nor excessive transport delay.

C.3.1.4 Digital Avionics Panel

C.3.1.5 Global Positioning System (GPS) navigator with moving map display.

C.3.1.6 Two-axis autopilot, and, as appropriate, a flight director (FD). This is only required when an autopilot is original standard equipment from the aircraft manufacturer.

C.3.1.7 Pitch trim (manual or electric pitch trim) permitting indicator movement either electrically or analog in an acceptable trim ratio (airplane only).

C.3.1.8 An independent visual system, panel, or screen that provides realistic cues in both day and night visual flight rules (VFR) and instrument flight rules (IFR) meteorological conditions to enhance a pilot's visual orientation in the vicinity of an airport including:

- Adjustable visibility parameters; and
- Adjustable ceiling parameters.

C.3.1.9 A fixed pilot seat appropriate to the aircraft configuration, including an adjustable height and an adjustable forward and aft seat position.

C.3.1.10 Rudder pedals secured to the cockpit floor structure, or that can be physically secured to the floor beneath the device in proper relation to cockpit orientation.

C.3.1.11 Push-to-talk switch on the control yoke.

C.3.1.12 A separate instructor station to permit effective interaction without interrupting the flight in overseeing the pilot's horizontal and vertical flight profiles in real time and space. This must include the ability to:

1. Oversee tracks along airways, holding entries and patterns, and Localizer (LOC) and glideslope (GS) alignment/deviation (or other approaches with a horizontal and vertical track).
2. Function as air traffic control (ATC) in providing vectors, etc., change in weather conditions, ceilings, visibilities, wind speed and direction, light/moderate/severe turbulence, and icing conditions.
3. Invoke failures in navigation and instruments, radio receivers, landing gear and flaps, engine power (partial and total), and other aircraft systems (pitot, electric, static, etc.) by using either a keyboard or mouse.

The Redbird CRV meets the following additional encouraged (not required) AATD CRITERIA:

C.3.2.1 Multi-panel or wrap-around visual system providing a 120 degrees or more of horizontal vision.

C.3.2.2 Automated ATC communications, scenario-based training (SBT), or line-oriented type training in which the instructor can evaluate pilot performance without having to act as ATC.

C.3.2.3 Simulated loss of performance and aerodynamic changes from ice accretion.

C.3.2.4 Realistic aircraft engine sound appropriate to the aircraft configuration, power settings, and speed.

C.3.2.5 A magnetic compass with incremental markings each 5 degrees, that displays the proper lead or lag during turns, and displays incremental markings typical of that shown in the aircraft.

SECTION 5: AIRCRAFT CONFIGURATION

List of Previously Approved Configurations (QAG v2.0A):

C208-G1



MTOW	FWD/AFT CG Limit
8750 lbs	199.15 in / 204.35 in

- **Yoke – Center (See Figures 2.1-2.8, 6.2)**
- **Throttle – Single Engine Complex Lever (T-P-M) (See Figure 3.1)**
- **Glass Cockpit G1000 (2)PFD/MFD & GFC700 Autopilot**

C208-S1



MTOW	FWD/AFT CG Limit
8750 lbs	199.15 in / 204.35 in

- **Yoke – Center (See Figures 2.1-2.8, 6.2)**
- **Throttle – Single Engine Complex Lever (T-P-M) (See Figure 3.1)**
- **Traditional analog gauges & 430/530 GPS with KFC225 Autopilot**

Additional configurations included in this version (QAG v 2.1B)

C208-G2



MTOW	FWD/AFT CG Limit
8807 lbs	199.15 in / 204.35 in

- **Yoke – Center (See Figures 2.1-2.8, 6.2)**
- **Throttle – Single Engine Complex Lever (T-P-M) (See Figure 3.1)**
- **Glass Cockpit G1000 (2)PFD/MFD & GFC700 Autopilot**

Performance Table

Aircraft Model	V_{SO}	V_{S1}	V_X	V_Y	V_A	V_{NE}	V_{MCA}	KTAS @ Cruise / 75% power setting*	Rate of Climb (fpm) @ (V_Y) / Full Power*	Single Engine Rate of Climb (V_{YSE})
Cessna Grand Caravan (208B)	48 KIAS	63 KIAS	70 KIAS	104 KIAS	148 KIAS	175 KIAS	N/A	160 KTAS**	975 fpm	N/A
							12,000' -->	159 KTAS	655 fpm	N/A
Cessna Grand Caravan EX (208B)	48 KIAS	63 KIAS	86 KIAS	108 KIAS	148 KIAS	175 KIAS	N/A	167 KTAS**	1331 fpm	N/A
							12,000' -->	166 KTAS	874 fpm	N/A

* Without cargo pod installed

** Performance @ 2,000’

SECTION 6: VISUAL SYSTEM WITH VFR, IFR, DAY, AND NIGHT CAPABILITY

Redbird CRV Visual System
The visual system is capable of providing a field-of-view of a minimum of 45 degrees horizontally and 30 degrees vertically, simultaneously for each pilot, including adjustable cloud base and visibility in night, dusk and day scenes.

- The Redbird CRV provides a means of recording the visual response time for the visual system that is installed.
- The Redbird CRV visual system is free of optical discontinuities and artifacts that create non-realistic cues.
- The visual system is directly displayed on six (6) or (8) LCD monitors inside the cockpit enclosure, situated in an arc around the Pilot. Each monitor is 28 cm tall, and 47 cm wide (36.5 cm x 63.5cm OPTIONAL). Based upon the designated Pilot Eye Point, these monitors provide a horizontal FOV of at least 220 (6 monitors) to 260 (8 monitors) degrees and a vertical FOV of minimally 30 degrees.

Daylight: The visual system provides full color presentations and sufficient surfaces with appropriate textural cues to conduct a visual approach, landing and airport movement. Surface shading effects are consistent with the simulated sun position.

Twilight: The visual system provides full color presentations of reduced ambient intensity, sufficient surfaces with appropriate textural cues that include self-illuminated objects such as road networks, ramp lighting and airport signage, to conduct a visual approach, landing and airport movement. Scenes include a definable horizon and typical terrain characteristics such as fields, roads and bodies of water and surfaces illuminated by representative ownship lighting.

Night: The visual system provides the same as above except the portrayal of reduced ambient intensity; therefore, there is no ground cues that are not self-illuminating or illuminated by ownship lights.

Designated Eye Point: The designated Pilot Eye Point is located 52 cm from the center of the forward most external view monitor, 61 cm from the left most external view monitor and 24 cm from the ceiling of the simulator enclosure. This point is roughly centered over the pilot's seat when it is adjusted to the forward most position, at a height consistent with the height of the pilot's head.

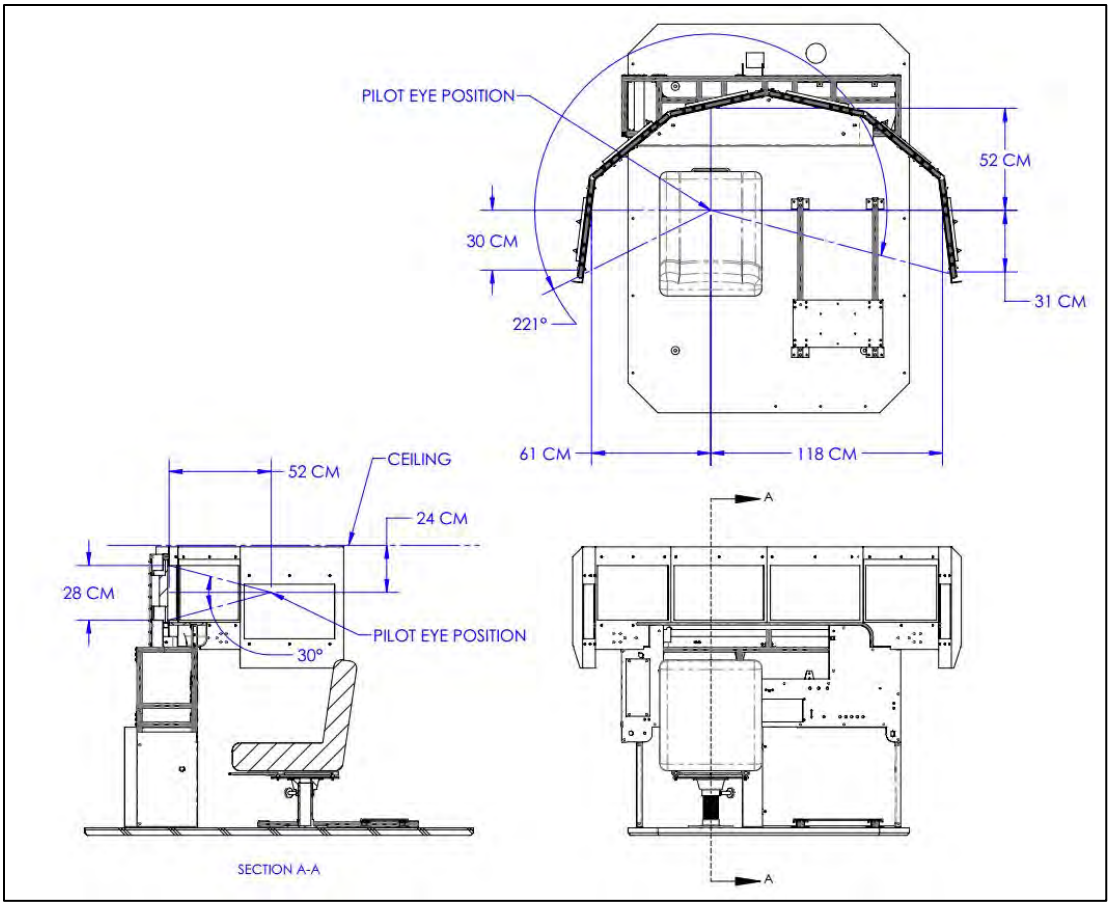


Figure 1: Designated Eye Point Diagram



Figure 6.3: Visual Displays – Type 1 (22” Display Monitors)

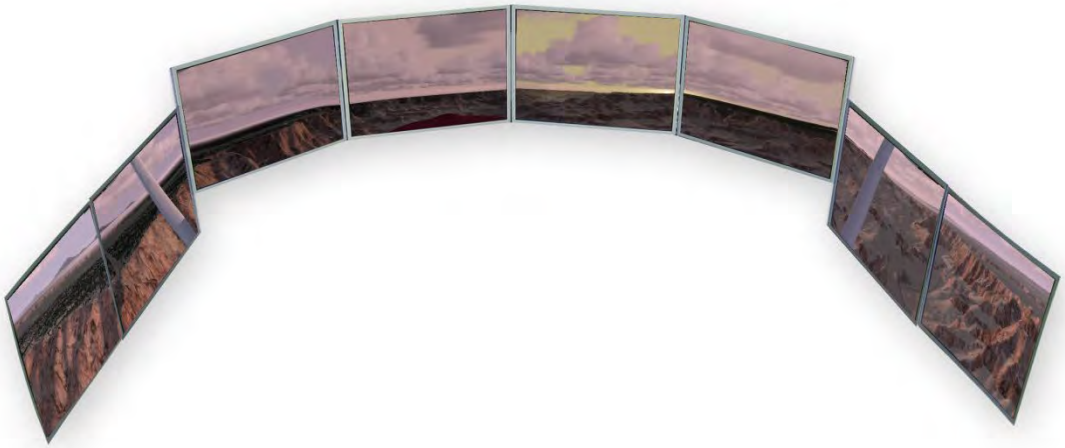


Figure 6.4: Visual Displays – Type 2 (22” Display Monitors)



Figure 6.7: Visual Displays – Type 3 (27” Display Monitors)



Figure 6.8: Visual Displays – Type 4 (27” Display Monitors)

SECTION 7: ATD FUNCTIONS & MANEUVERS CHECKLIST

PROCEDURES AND TASKS TEST CHECKLIST

Maneuvers and Tasks	Yes/No/NA
a) Pretakeoff	
1) Engine start	Yes
2) Taxi and brake operation	Yes
b) Takeoff	
1) AIRPLANE Takeoff	
i) Run-up and powerplant checks	Yes
ii) Acceleration characteristics	Yes
iii) Nosewheel and rudder steering	Yes
iv) Effect of crosswind	Yes
v) Instrument	Yes
vi) Landing gear, wing flap operation	Yes
2) HELICOPTER Takeoff	
i) Powerplant checks	NA
ii) From hover	NA
iii) From ground	NA
Maneuvers and Tasks	**Yes/No/NA**
iv) Vertical	NA
v) Running	NA
c) In-Flight Operation	
1) AIRPLANE In-Flight Operation	
i) Climb	Yes
(a) Normal and max. performance	Yes
(b) One-engine-inoperative procedures (multiengine)	Yes
ii) Cruise	Yes
(a) Performance characteristics (speed vs. power)	Yes
(b) Normal and steep turns	Yes
(c) Approach to stalls (i.e., stall warning), stalls, and recovery. Execute from takeoff, cruise, and approach and landing configurations.	Yes
(d) In-flight engine shutdown (multiengine)	Yes
(e) Fuel selector function	Yes
(f) In-flight engine start	Yes
iii) Approach	Yes
(a) Normal (with and without flaps) (check gear warning, if applicable)	Yes
(b) Best glide no power	Yes
iv) Landings	Yes
2) HELICOPTER In-Flight Operation	
i) Hovering and air taxi	NA

(a) Forward	NA
(b) Rearward	NA
(c) Sideward	NA
(d) Turns	NA
ii) Climb	NA
iii) Cruise	NA
(a) Performance characteristics (speed vs. power)	NA
Maneuvers and Tasks	**Yes/No/NA**
(b) Turns	NA
(i) Recovery	NA
(ii) Skidding	NA
(iii) Slipping	NA
(iv) Steep turns	NA
(c) In-flight engine shutdown and start (multiengine)	NA
(d) Descents	NA
(e) Straight in and 180º autorotation	NA
(f) Landings	NA
d) Instrument Approaches	
1) Nonprecision	
i) GPS and LPV	Yes
ii) GPS-WAAS (optional)	Yes
iii) All engines operating	Yes
iv) One or more engines inoperative	Yes
v) Approach procedures (VOR, VOR/DME, LOC procedures on an ILS, LDA, RNAV (RDP) or RNAV (GPS) to LNAV, LNAV/VNAV or LPV)	Yes
2) Precision	
i) ILS	Yes
ii) GLS (optional)	No
iii) Effects of crosswind	Yes
iv) With engine inoperative (multiengine)	Yes
v) Missed approach	Yes
(a) Normal	Yes
(b) With engine(s) inoperative (multiengine)	Yes
e) Surface Operations	
1) AIRPLANE Surface Operations (Post Landing)	
i) Approach and landing roll	Yes
ii) Braking operation	Yes
Maneuvers and Tasks	**Yes/No/NA**
iii) Reverse thrust operation, if applicable	NA
2) HELICOPTER Surface Operations	
i) Landings	NA
ii) Landing area operations	NA
f) HELICOPTER Emergency Operations	

1) Power failure at hover	NA
2) Power failure at altitude	NA
3) System and equipment malfunctions	NA
4) Settling with power (optional)	NA
5) Low rotor RPM recovery (optional)	NA
6) Antitorque system failure	NA
7) Dynamic rollover (optional)	NA
g) Any Flight Phase	
1) Aircraft and Powerplant Systems	Yes
i) Electrical, mechanical, or hydraulic	Yes
ii) Flaps (airplane)	Yes
iii) Fuel selector and oil temp/pressure	Yes
iv) Landing gear (if applicable)	Yes
2) Flight Management and Guidance Systems	Yes
i) Autopilot (if standard equipment)	Yes
ii) Flight director (AATD only)/system displays (if installed)	Yes
iii) Navigation systems	Yes
iv) Stall warning systems avoidance (airplane)	Yes
v) Multi-function displays (if applicable)	Varies* (Based on Configuration)
3) Airborne Procedures	Yes
i) Holding	Yes
ii) Uncoordinated turns – slipping and skidding demo	Yes
Maneuvers and Tasks	**Yes/No/NA**
iii) Configuration and power changes and resulting pitch changes	Yes
iv) Compass turns and appropriate errors (if installed)	Yes
4) Engine Shutdown and Parking	Yes
i) Systems operation	Yes
ii) Parking brake operation (if installed) (airplane)	Yes
h) Can simulate engine failure, including failures due to simulated loss of oil pressure or fuel starvation.	Yes (Except oil pressure loss)
i) Can simulate the following equipment or system failures:	
1) Alternator or generator failure.	Yes
2) Vacuum pump/pressure failure and the associated flight instrument failures.	Yes
3) Gyroscopic flight instrument failures.	Yes
4) Pitot/static system malfunction and the associated flight instrument failures.	Yes
5) Electronic flight deck display malfunctions.	Yes
6) Landing gear (if retractable) or flap malfunctions.	Yes
j) Independent Instructor Station Requirements (AATD Only)	
1) Displays published airways and holding patterns.	Yes

2) Displays aircraft position and track.	Yes
3) Displays aircraft altitude and speed.	Yes
4) Displays NAVAIDs and airports.	Yes
5) Can record and replay aircraft ground track history for entire training session.	Yes
6) Can invoke instrument or equipment failures.	Yes

Any changes or modifications to this device which have not been reviewed, evaluated, and approved in writing by General Aviation and Commercial Division, AFS-800 may terminate FAA approval of this aviation training device.